

How to use generalised eigenvector to find a solution of system of differential equations (SDE) for degenerate node?

Answer:

For a 2x2 system $\mathbf{x}' = \mathbf{A}\mathbf{x}$, if we find a repeated eigenvalue λ but can only find one independent eigenvector $\mathbf{v}$, we can form one solution:

$$\mathbf{x}_1(t) = \mathbf{v}e^{\lambda t}$$

We need a second, linearly independent solution, $\mathbf{x}_2(t)$. We cannot simply multiply $\mathbf{x}_1(t)$ by t , as we might in a second-order scalar ODE. The correct form of the second solution involves both the eigenvector $\mathbf{v}$ and a new vector $\mathbf{w}$, the generalized eigenvector associated with $\mathbf{v}$.

The second solution takes the form:

$$\mathbf{x}_2(t) = (\mathbf{v}t + \mathbf{w})e^{\lambda t}$$

How to Find the Generalized Eigenvector

A generalized eigenvector $\mathbf{w}$ is defined by the following relationship with its corresponding eigenvector $\mathbf{v}$ and eigenvalue λ :

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{w} = \mathbf{v}$$

So we can find $\mathbf{w}$ by solving the non-homogeneous system with $(\mathbf{A} - \lambda\mathbf{I})$ as the coefficient matrix, and $\mathbf{v}$ as the constant term. Note that we cannot replace $(\mathbf{A} - \lambda\mathbf{I})$ by $(\lambda\mathbf{I} - \mathbf{A})$ here.

Step-by-Step Guide to Solving a 2x2 System with a Degenerate Node

2. **Find the Eigenvalues:** You will find a single, repeated real eigenvalue, λ .
3. **Find the Eigenvector:** You will find a single linearly independent eigenvector, $\mathbf{v}$. This gives you your first solution: $\mathbf{x}_1(t) = \mathbf{v}e^{\lambda t}$.
4. **Find the Generalized Eigenvector:** Using the λ and $\mathbf{v}$ you just found, solve the system $(\mathbf{A} - \lambda\mathbf{I})\mathbf{w} = \mathbf{v}$ for the generalized eigenvector $\mathbf{w}$. This gives you your second solution: $\mathbf{x}_2(t) = (\mathbf{v}t + \mathbf{w})e^{\lambda t}$.
5. **Construct the General Solution:** Combine the two parts to form the general solution:

$$\mathbf{x}(t) = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t) \quad \mathbf{x}(t) = c_1\mathbf{v}e^{\lambda t} + c_2(\mathbf{v}t + \mathbf{w})e^{\lambda t}$$

Scenario 1

Let's solve the system $\mathbf{x}' = \mathbf{A}\mathbf{x}$ for the matrix $\mathbf{A} = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$

Step 1: Find the Eigenvalues (either by solving the characteristic polynomial or use MATLAB). We have a repeated eigenvalue: $\lambda = 2$.

Step 2: Find the Eigenvector (either by solving the homogeneous system $(2\mathbf{I} - \mathbf{A})\mathbf{v} = \mathbf{0}$ or use MATLAB).

We have a linearly independent eigenvector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ (or any of its scalar multiples).

Our first solution is $\mathbf{x}_1(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t}$.

Step 3: Find the Generalized Eigenvector

We solve $(\mathbf{A} - 2\mathbf{I})\mathbf{w} = \mathbf{v}$ for $\mathbf{w}$:

$$\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Both rows give the equation $w_1 - w_2 = 1$. The general solution is $w_1 = 1 + s$ and $w_2 = s$ where s is a parameter. We can choose s to be any value to get a generalised eigenvector $\mathbf{w}$. For example, setting $s = 0$ gives $\mathbf{w} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

Step 4: Construct the General Solution

Now we assemble the general solution $\mathbf{x}(t) = c_1 \mathbf{v} e^{\lambda t} + c_2 (\mathbf{v}t + \mathbf{w}) e^{\lambda t}$ using $\lambda=2$, $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and $\mathbf{w} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ to get

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t} + c_2 \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) e^{2t}$$

We can combine the terms for c_2 :

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} t+1 \\ t \end{pmatrix} e^{2t} \quad (*)$$

In step 3, it is also possible to let s be any other value, say $s = 1$ and get an alternative generalised eigenvector $\mathbf{w} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ (or another vector correspondingly).

In this case, the general solution $\mathbf{x}(t) = c_1 \mathbf{v} e^{\lambda t} + c_2 (\mathbf{v}t + \mathbf{w}) e^{\lambda t}$ becomes

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t} + c_2 \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} t + \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right) e^{2t}$$

or more concisely

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} t+2 \\ t+1 \end{pmatrix} e^{2t} \quad (**)$$

The two general solutions $(*)$ and $(**)$ are equivalent to each other.